1.57 Equations of Planes - Cartesian Equations

Worked Solutions

1 a)

$$2 \times (3) + 8 \times (1) - 3 \times \left(\frac{10}{3}\right) = 6 + 8 - 10 = 4.$$

So $\left(3, 1, \frac{10}{3}\right)$ lies on the plane Π.

1 b)

$$2 \times (21) + 8 \times (-2) - 3 \times (7) = 42 - 16 - 21 = 5 \neq 4.$$

So $(21, -2, 7)$ does not lie on the plane Π.

2 a)

$$2(3) + (-5) + 6(4) = 6 - 5 + 24 = 25 \neq 24.$$

So A does not lie on the plane Π_1.

2 b)

$$-5(3) - 2(-5) + 3(4) = -15 + 10 + 12 = 7.$$

So A does lie on the plane Π_2.

2 c)

$$8(3) + 3(-5) - 8(4) = 24 - 15 - 32 = -23.$$

So A does lie on the plane Π_3.

2 d)

$$\frac{1}{2}(3) + \frac{3}{2}(-5) + 3(4) = \frac{3}{2} - \frac{15}{2} + 12 = 6.$$

So A does lie on the plane Π_4.

3 a)

$$\begin{pmatrix} 3 \\ -5 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -16 \\ -7 \\ 13 \end{pmatrix} = 3 \times (-16) + (-5) \times (-7) + 1 \times 13 = 0.$$

So the plane Π contains the vector $\begin{pmatrix} -16 \\ -7 \\ 13 \end{pmatrix}$.

3 b)

$$\begin{pmatrix} 3 \\ -5 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 13 \\ 4 \\ -19 \end{pmatrix} = 3 \times 13 + (-5) \times 4 + 1 \times (-19) = 0.$$

So the plane Π contains the vector $\begin{pmatrix} 13 \\ 4 \\ -19 \end{pmatrix}$.

3 c)

$$\begin{pmatrix} 3 \\ -5 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -13 \\ -7 \\ -16 \end{pmatrix} = 3 \times (-13) + (-5) \times (-7) + 1 \times (-16) = -20 \neq 0.$$

So the plane Π does not contain the vector $\begin{pmatrix} -13 \\ -7 \\ -16 \end{pmatrix}$.

4 a)

The plane has equation $2x + 2y + z = d$.

$$d = 2(1) + 2(4) + 1 = 11.$$

$$\therefore 2x + 2y + z = 11.$$

4 b)

The plane has equation $-x + 4y - 3z = d$.

$$d = -(6) + 4(1) - 3(-2) = 4.$$

$$\therefore -x + 4y - 3z = 4.$$

4 c)

The plane has equation $5x - y - 2z = d$.

$$d = 5(-3) - (-4) - 2(-9) = 7.$$

$$\therefore 5x - y - 2z = 7.$$

4 d)

The plane has equation $-2x + 3y + 10z = d$.

$$d = -2\left(\frac{1}{2}\right) + 3\left(-\frac{3}{2}\right) + 10\left(\frac{5}{2}\right) = \frac{39}{2}.$$

$$\therefore -2x + 3y + 10z = \frac{39}{2} \quad \text{or} \quad -4x + 6y + 20z = 39.$$

4 e)

The plane has equation $0x + 7y - 8z = d$.

$$d = 0(3) + 7(0) - 8(-2) = 16.$$

$$\therefore 7y - 8z = 16.$$

4 f)

The plane has equation $-4x + 15y + 0z = d$.

$$d = -4\left(-\frac{1}{3}\right) + 15\left(\frac{1}{4}\right)0\left(\frac{2}{3}\right) = \frac{61}{12}.$$

$$\therefore -4x + 15y = \frac{61}{12} \quad \text{or} \quad -48x + 180y = 61.$$

5 a)

The plane has equation $5x - 2y + 4z = d$.

$$d = 5(2) - 2(1) + 4(3) = 20.$$

$$\therefore 5x - 2y + 4z = 20.$$

5 b)

$B(3, 1, -5)$:

$$5(3) - 2(1) + 4(-5) = -7 \neq 20.$$

So B does not lie on the plane. $C(-2, 0, 8)$:

$$5(-2) - 2(0) + 4(8) = 22 \neq 20.$$

So C does not lie on the plane.

6)

The plane has equation $4x - y + z = d$.

$$d = 4(1) - (1) + (4) = 7.$$

$$\therefore 4x - y + z = 7.$$

7)

The plane has equation $-3x + 8y - z = d$.

$$d = -3(2) + 8\left(\frac{1}{2}\right) - (-8) = 6.$$

$$\therefore -3x + 8y - z = 6.$$

8)

The plane has equation $kx + 3y - 2kz = d$.

$$d = k(3) + 3(-4k) - 2k(5) = 3k - 12k - 10k = -19k.$$

$$\therefore kx + 3y - 2kz = -19k.$$

9 a)

$$\begin{pmatrix}-1\\6\\-2\end{pmatrix}\cdot\begin{pmatrix}1\\1\\p\end{pmatrix}=0\Rightarrow -1\times 1+6\times 1+(-2)\times p=0\Rightarrow 5-2p=0\Rightarrow p=\frac{5}{2}.$$

9 b)

The plane has equation $x+y+\frac{5}{2}z=d$.

$$d=1(2)+1(1)+\frac{5}{2}(1)=\frac{11}{2}.$$

$$\therefore x+y+\frac{5}{2}z=\frac{11}{2}.$$

$$\Rightarrow 2x+2y+5z=11.$$

10)

$$\begin{pmatrix}k\\-6\\1\end{pmatrix}\cdot\begin{pmatrix}4\\8\\-3\end{pmatrix}=0\Rightarrow k\times 4+(-6)\times 8+1\times(-3)=0.$$

$$\Rightarrow 4k-48-3=0\Rightarrow 4k-51=0\Rightarrow k=\frac{51}{4}.$$

11)

$$\begin{pmatrix}5\\1\\-k\end{pmatrix}\cdot\begin{pmatrix}2k\\-7\\2\end{pmatrix}=0\Rightarrow 5\times(2k)+1\times(-7)+(-k)\times 2=0.$$

$$\Rightarrow 10k-7-2k=0\Rightarrow 8k-7=0\Rightarrow k=\frac{7}{8}.$$

12)

$$\begin{pmatrix}k+1\\-2\\3k\end{pmatrix}\cdot\begin{pmatrix}4k\\-(k+2)\\-10\end{pmatrix}=0\Rightarrow (k+1)\times 4k+(-2)\times(-(k+2))+3k\times(-10)=0.$$

$$\Rightarrow 4k^2+4k+2k+4-30k=0.$$

$$\Rightarrow 4k^2-24k+4=0.$$

$$\Rightarrow k^2-6k+1=0.$$

$$\Rightarrow (k-3)^2-9+1=0.$$

$$\Rightarrow (k-3)^2=8.$$

$$\Rightarrow k=3\pm 2\sqrt{2}.$$

13 a)

First find the angle between the vector and the normal to the plane:

$$\cos\theta = \frac{\begin{pmatrix}1\\2\\4\end{pmatrix}\cdot\begin{pmatrix}1\\4\\3\end{pmatrix}}{\sqrt{1^2+2^2+4^2}\sqrt{1^2+4^2+3^2}}.$$

$$\Rightarrow \cos\theta = \frac{1\times 1+2\times 4+4\times 3}{\sqrt{21}\times\sqrt{26}}.$$

$$\Rightarrow \cos\theta = \frac{\sqrt{546}}{26} \Rightarrow \theta = 26.010\ldots^\circ.$$

So angle between the vector and the plane is: Angle $= 90^\circ - 26.010\ldots^\circ = 64.0^\circ$ (1 DP).

13 b)

First find the angle between the vector and the normal to the plane:

$$\cos\theta = \frac{\begin{pmatrix}-1\\3\\-8\end{pmatrix}\cdot\begin{pmatrix}2\\-1\\2\end{pmatrix}}{\sqrt{(-1)^2+3^2+(-8)^2}\sqrt{2^2+(-1)^2+2^2}}.$$

$$\Rightarrow \cos\theta = \frac{(-1)\times 2+3\times(-1)+(-8)\times 2}{\sqrt{74}\times\sqrt{9}}.$$

$$\Rightarrow \cos\theta = -\frac{7\sqrt{74}}{74} \Rightarrow \theta = 144.462\ldots^\circ.$$

Acute Angle $= 180^\circ - 144.462\ldots^\circ = 35.537\ldots^\circ$. So angle between the vector and the plane is: Angle $= 90^\circ - 35.537\ldots^\circ = 54.5^\circ$ (1 DP).

13 c)

First find the angle between the vector and the normal to the plane:

$$\cos\theta = \frac{\begin{pmatrix}1\\8\\0\end{pmatrix}\cdot\begin{pmatrix}5\\0\\-4\end{pmatrix}}{\sqrt{1^2+8^2+0^2}\sqrt{5^2+0^2+(-4)^2}}.$$

$$\Rightarrow \cos\theta = \frac{1\times 5+8\times 0+0\times(-4)}{\sqrt{65}\times\sqrt{41}}.$$

$$\Rightarrow \cos\theta = \frac{\sqrt{2665}}{533} \Rightarrow \theta = 84.441\ldots^\circ.$$

So angle between the vector and the plane is: Angle $= 90^\circ - 84.441\ldots^\circ = 5.6^\circ$ (1 DP).

13 d)

First find the angle between the vector and the normal to the plane:

$$\cos\theta = \frac{\begin{pmatrix}9\\0\\-3\end{pmatrix}\cdot\begin{pmatrix}-1\\3\\2\end{pmatrix}}{\sqrt{9^2+0^2+(-3)^2}\sqrt{(-1)^2+3^2+2^2}}.$$

$$\Rightarrow \cos\theta = \frac{9 \times (-1) + 0 \times 3 + (-3) \times 2}{\sqrt{90} \times \sqrt{14}}.$$

$$\Rightarrow \cos\theta = -\frac{\sqrt{35}}{14} \Rightarrow \theta = 114.997\ldots^\circ.$$

Acute Angle $= 180^\circ - 114.997\ldots^\circ = 65.002\ldots^\circ$. So angle between the vector and the plane is: Angle $= 90^\circ - 65.002\ldots^\circ = 25.0^\circ$ (1 DP).

14 a)

The XY plane has equation $z = 0$.

$$\cos\theta = \frac{\begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}{\sqrt{1^2 + (-1)^2 + (-2)^2} \times \sqrt{0^2 + 0^2 + 1^2}}.$$

$$\Rightarrow \cos\theta = \frac{1 \times 0 + (-1) \times 0 + (-2) \times 1}{\sqrt{6} \times \sqrt{1}}.$$

$$\Rightarrow \cos\theta = -\frac{2}{\sqrt{6}} \Rightarrow \theta = 144.735\ldots^\circ.$$

Acute Angle $= 180^\circ - 144.735\ldots^\circ = 35.3^\circ$ (1 DP).

14 b)

The XZ plane has equation $y = 0$.

$$\cos\theta = \frac{\begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}}{\sqrt{1^2 + (-1)^2 + (-2)^2} \times \sqrt{0^2 + 1^2 + 0^2}}.$$

$$\Rightarrow \cos\theta = \frac{1 \times 0 + (-1) \times 1 + (-2) \times 0}{\sqrt{6} \times \sqrt{1}}.$$

$$\Rightarrow \cos\theta = -\frac{1}{\sqrt{6}} \Rightarrow \theta = 114.094\ldots^\circ.$$

Acute Angle $= 180^\circ - 114.094\ldots^\circ = 65.9^\circ$ (1 DP).

14 c)

The YZ plane has equation $x = 0$.

$$\cos\theta = \frac{\begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}{\sqrt{1^2 + (-1)^2 + (-2)^2} \times \sqrt{0^2 + 1^2 + 0^2}}.$$

$$\Rightarrow \cos\theta = \frac{1 \times 1 + (-1) \times 0 + (-2) \times 0}{\sqrt{6} \times \sqrt{1}}.$$

$$\Rightarrow \cos\theta = \frac{1}{\sqrt{6}} \Rightarrow \theta = 65.905\ldots^\circ.$$

Acute Angle $= 65.9^\circ$ (1 DP).

15)

$$\cos\theta = \frac{\begin{pmatrix}2\\3\\-8\end{pmatrix}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}}{\sqrt{2^2+3^2+(-8)^2}\times\sqrt{0^2+0^2+1^2}}.$$
$$\Rightarrow \cos\theta = \frac{2\times 0+3\times 0+(-8)\times 1}{\sqrt{77}\times\sqrt{1}}.$$
$$\Rightarrow \cos\theta = -\frac{8}{\sqrt{77}} \Rightarrow \theta = 155.739\ldots^\circ.$$

Acute Angle $= 24.260\ldots^\circ = 24.3^\circ$ (1 DP). A 20% incline is an angle $\phi = \arctan\frac{20}{100} = 11.3^\circ$ (1 DP). $24.3^\circ > 11.3^\circ$ so the new road is too steep and will be rejected.

16 a)

$$\cos\theta = \frac{\begin{pmatrix}1\\1\\-1\end{pmatrix}\cdot\begin{pmatrix}1\\1\\-2\end{pmatrix}}{\sqrt{1^2+1^2+(-1)^2}\times\sqrt{1^2+1^2+(-2)^2}}.$$
$$\Rightarrow \cos\theta = \frac{1\times 1+1\times 1+(-1)\times(-2)}{\sqrt{3}\times\sqrt{6}}.$$
$$\Rightarrow \cos\theta = \frac{4}{3\sqrt{2}} = \frac{2\sqrt{2}}{3} \Rightarrow \theta = 19.471\ldots^\circ.$$

Acute Angle $= 19.5^\circ$ (1 DP).

16 b)

$$\cos\theta = \frac{\begin{pmatrix}3\\2\\1\end{pmatrix}\cdot\begin{pmatrix}8\\1\\2\end{pmatrix}}{\sqrt{3^2+2^2+1^2}\times\sqrt{8^2+1^2+2^2}}.$$
$$\Rightarrow \cos\theta = \frac{3\times 8+2\times 1+1\times 2}{\sqrt{14}\times\sqrt{69}}.$$
$$\Rightarrow \cos\theta = \frac{28}{\sqrt{966}} = \frac{2\sqrt{966}}{69} \Rightarrow \theta = 25.725\ldots^\circ.$$

Acute Angle $= 25.7^\circ$ (1 DP).

16 c)

$$\cos\theta = \frac{\begin{pmatrix}4\\-1\\-1\end{pmatrix}\cdot\begin{pmatrix}-1\\1\\1\end{pmatrix}}{\sqrt{4^2+(-1)^2+(-1)^2}\times\sqrt{(-1)^2+1^2+1^2}}.$$
$$\Rightarrow \cos\theta = \frac{4\times(-1)+(-1)\times 1+(-1)\times 1}{\sqrt{18}\times\sqrt{3}}.$$
$$\Rightarrow \cos\theta = -\frac{6}{3\sqrt{6}} = -\frac{\sqrt{6}}{3} \Rightarrow \theta = 144.735\ldots^\circ.$$

Acute Angle $= 180^\circ - 144.735\ldots^\circ = 35.3^\circ$ (1 DP).

16 d)

$$\cos\theta = \frac{\begin{pmatrix}-5\\-2\\1\end{pmatrix}\cdot\begin{pmatrix}9\\10\\-6\end{pmatrix}}{\sqrt{(-5)^2+(-2)^2+1^2}\times\sqrt{9^2+10^2+(-6)^2}}.$$

$$\Rightarrow \cos\theta = \frac{(-5)\times 9+(-2)\times 10+1\times(-6)}{\sqrt{30}\times\sqrt{217}}.$$

$$\Rightarrow \cos\theta = -\frac{71}{\sqrt{6510}} = -\frac{71\sqrt{6510}}{6510} \Rightarrow \theta = 151.638\ldots^\circ.$$

Acute Angle $= 180^\circ - 151.638\ldots^\circ = 28.4^\circ$ (1 DP).

16 e)

$$\cos\theta = \frac{\begin{pmatrix}1\\0\\-4\end{pmatrix}\cdot\begin{pmatrix}0\\1\\11\end{pmatrix}}{\sqrt{1^2+0^2+(-4)^2}\times\sqrt{0^2+1^2+11^2}}.$$

$$\Rightarrow \cos\theta = \frac{1\times 0+0\times 1+(-4)\times 11}{\sqrt{17}\times\sqrt{122}}.$$

$$\Rightarrow \cos\theta = -\frac{44}{\sqrt{2074}} = -\frac{22\sqrt{2074}}{1037} \Rightarrow \theta = 165.051\ldots^\circ.$$

Acute Angle $= 180^\circ - 165.051\ldots^\circ = 14.9^\circ$ (1 DP).

16 f)

$$\cos\theta = \frac{\begin{pmatrix}\sqrt{2}\\\sqrt{6}\\0\end{pmatrix}\cdot\begin{pmatrix}2\sqrt{2}\\2\sqrt{3}\\-1\end{pmatrix}}{\sqrt{(\sqrt{2})^2+(\sqrt{6})^2+0^2}\times\sqrt{(2\sqrt{2})^2+(2\sqrt{3})^2+(-1)^2}}.$$

$$\Rightarrow \cos\theta = \frac{\sqrt{2}\times 2\sqrt{2}+\sqrt{6}\times 2\sqrt{3}+0\times(-1)}{\sqrt{8}\times\sqrt{21}}.$$

$$\Rightarrow \cos\theta = \frac{4+6\sqrt{2}}{2\sqrt{42}} = \frac{\sqrt{42}+3\sqrt{21}}{21} \Rightarrow \theta = 15.579\ldots^\circ.$$

Acute Angle $= 15.6^\circ$ (1 DP).

17)

$$\frac{\begin{pmatrix}1\\12\\0\end{pmatrix}\cdot\begin{pmatrix}1\\6\\k\end{pmatrix}}{\sqrt{1^2+12^2+0^2}\times\sqrt{1^2+6^2+k^2}} > \cos 5^\circ = 0.996\ldots.$$

$$\Rightarrow \frac{1\times 1+12\times 6+0\times k}{\sqrt{145}\times\sqrt{k^2+37}} > \cos 5^\circ.$$

$$\Rightarrow \frac{73}{\sqrt{145k^2+5365}} > \cos 5^\circ.$$

$$\Rightarrow \frac{73}{\cos 5^\circ} > \sqrt{145k^2 + 5365}.$$
$$\Rightarrow \frac{5329}{\cos^2 5^\circ} > 145k^2 + 5365.$$
$$\Rightarrow k^2 < \frac{5329}{145\cos^2 5^\circ} - 37 = 0.0330316\ldots.$$
$$\Rightarrow -0.182 < k < 0.182.$$

18)

$$\frac{\begin{pmatrix} k \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 20 \\ 39 \end{pmatrix}}{\sqrt{k^2 + 1^2 + 2^2} \times \sqrt{0^2 + 20^2 + 39^2}} < \cos 1^\circ = 0.999847\ldots.$$
$$\Rightarrow \frac{k \times 0 + 1 \times 20 + 2 \times 39}{\sqrt{k^2 + 5} \times \sqrt{1921}} < \cos 1^\circ.$$
$$\Rightarrow \frac{98}{\sqrt{1921k^2 + 9605}} < \cos 1^\circ.$$
$$\Rightarrow \frac{98}{\cos 1^\circ} < \sqrt{1921k^2 + 9605}.$$
$$\Rightarrow \frac{9604}{\cos^2 1^\circ} < 1921k^2 + 9605.$$
$$\Rightarrow k^2 > \frac{9604}{1921\cos^2 1^\circ} - 5 = 0.001002675647\ldots.$$
$$\Rightarrow k < -0.0317 \text{ (3 SF) or } k > 0.0317 \text{ (3 SF)}.$$

19)

$$\cos 45^\circ = \frac{\begin{pmatrix} 6 \\ -1 \\ k \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -3 \\ 0 \end{pmatrix}}{\sqrt{6^2 + (-1)^2 + k^2} \times \sqrt{3^2 + (-3)^2 + 0^2}}.$$
$$\Rightarrow \frac{\sqrt{2}}{2} = \frac{6 \times 3 + (-1) \times (-3) + k \times 0}{\sqrt{37 + k^2} \times \sqrt{18}}.$$
$$\Rightarrow \frac{\sqrt{2}}{2} = \frac{21}{\sqrt{666 + 18k^2}}.$$
$$\Rightarrow \sqrt{1332 + 36k^2} = 42.$$
$$\Rightarrow 1332 + 36k^2 = 1764.$$
$$\Rightarrow k^2 = 12 \Rightarrow k = \pm 2\sqrt{3}.$$

As k is a positive constant, $k = 2\sqrt{3}$.

20)

$$\cos 30^\circ = \frac{\begin{pmatrix} 9 \\ k \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix}}{\sqrt{9^2 + k^2 + (-3)^2} \times \sqrt{2^2 + 0^2 + (-2)^2}}.$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{9 \times 2 + k \times 0 + (-3) \times (-2)}{\sqrt{90 + k^2} \times \sqrt{8}}.$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{24}{\sqrt{720 + 8k^2}}.$$

$$\Rightarrow \sqrt{2160 + 24k^2} = 48.$$

$$\Rightarrow 2160 + 24k^2 = 2304.$$

$$\Rightarrow k^2 = 6 \Rightarrow k = \pm\sqrt{6}.$$

As k is a positive constant, $k = \sqrt{6}$.

21)

$$\cos 60^\circ = \frac{\begin{pmatrix} 8 \\ 1 \\ 2k \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}}{\sqrt{8^2 + 1^2 + (2k)^2} \times \sqrt{4^2 + (-2)^2 + 0^2}}.$$

$$\Rightarrow \frac{1}{2} = \frac{8 \times 4 + 1 \times (-2) + 2k \times 0}{\sqrt{4k^2 + 65} \times \sqrt{20}}.$$

$$\Rightarrow \frac{1}{2} = \frac{30}{\sqrt{80k^2 + 1300}}.$$

$$\Rightarrow \sqrt{80k^2 + 1300} = 60.$$

$$\Rightarrow 80k^2 + 1300 = 3600.$$

$$\Rightarrow k^2 = \frac{115}{4} \Rightarrow k = \pm\frac{\sqrt{115}}{2}.$$

As k is a negative constant, $k = -\frac{\sqrt{115}}{2}$.

22)

$$\frac{1}{9} = \frac{\begin{pmatrix} 8 \\ -4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ k \\ -2 \end{pmatrix}}{\sqrt{8^2 + (-4)^2 + 1^2} \times \sqrt{4^2 + k^2 + (-2)^2}}.$$

$$\Rightarrow \frac{1}{9} = \frac{8 \times 4 + (-4) \times k + 1 \times (-2)}{9 \times \sqrt{20 + k^2}}.$$

$$\Rightarrow 1 = \frac{30 - 4k}{\sqrt{20 + k^2}}.$$

$$\Rightarrow \sqrt{20 + k^2} = 30 - 4k.$$

$$\Rightarrow 20 + k^2 = (30 - 4k)^2.$$

$$\Rightarrow 20 + k^2 = 900 - 240k + 16k^2.$$
$$\Rightarrow 0 = 15k^2 - 240k + 880.$$
$$\Rightarrow 0 = 3k^2 - 48k + 176.$$
$$\Rightarrow 0 = 3(k^2 - 16k) + 176.$$
$$\Rightarrow 0 = 3((k-8)^2 - 64) + 176.$$
$$\Rightarrow 0 = 3(k-8)^2 - 192 + 176.$$
$$\Rightarrow 0 = 3(k-8)^2 - 16.$$
$$\Rightarrow (k-8)^2 = \frac{16}{3}.$$
$$\Rightarrow k - 8 = \pm\frac{4}{\sqrt{3}}.$$
$$\Rightarrow k = 8 \pm \frac{4}{\sqrt{3}}.$$

Solution must solve $\sqrt{20 + k^2} = 30 - 4k$. If $k = 8 + \frac{4}{\sqrt{3}}$ then $30 - 4k = -\frac{6+16\sqrt{3}}{3} < 0$. If $k = 8 - \frac{4}{\sqrt{3}}$ then $30 - 4k = \frac{-6+16\sqrt{3}}{3} > 0$.

$$\therefore k = 8 - \frac{4}{\sqrt{3}}.$$

23)

Π_1 has equation $10x - 5y + pz = d_1$. Π_2 has equation $4x + 4y - 8z = d_2$. $(2, 1, -5)$ lies on Π_1 so:

$$10(2) - 5(1) + p(-5) = d_1 \Rightarrow 15 - 5p = d_1.$$

The angle between Π_1 and Π_2 is $60°$ so:

$$\cos 60° = \frac{\begin{pmatrix} 10 \\ -5 \\ p \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 4 \\ -8 \end{pmatrix}}{\sqrt{10^2 + (-5)^2 + p^2} \times \sqrt{4^2 + 4^2 + (-8)^2}}.$$
$$\Rightarrow \frac{1}{2} = \frac{10 \times 4 + (-5) \times 4 + p \times (-8)}{\sqrt{125 + p^2} \times \sqrt{96}}.$$
$$\Rightarrow \frac{1}{2} = \frac{20 - 8p}{\sqrt{12000 + 96p^2}}.$$
$$\Rightarrow \sqrt{12000 + 96p^2} = 40 - 16p.$$
$$\Rightarrow 12000 + 96p^2 = (40 - 16p)^2.$$
$$\Rightarrow 12000 + 96p^2 = 1600 - 1280p + 256p^2.$$
$$\Rightarrow 0 = 160p^2 - 1280p - 10400.$$
$$\Rightarrow 0 = p^2 - 8p - 65.$$
$$\Rightarrow 0 = (p - 13)(p + 5).$$
$$\Rightarrow p = 13 \text{ or } p = -5.$$

As p is a negative constant, $p = -5$.

$$\Rightarrow d_1 = 15 - 5(-5) = 40.$$

$$\therefore \text{The cartesian equation of } \Pi_1 \text{ is } 10x - 5y - 5z = 40 \Rightarrow 2x - y - z = 8.$$

24)

If $\mathbf{m}$ lies in the plane, then

$$\begin{pmatrix}3\\0\\1\end{pmatrix}\cdot\begin{pmatrix}p\\q\\r\end{pmatrix}=0\Rightarrow 3p+r=0\Rightarrow r=-3p.$$

$$|\mathbf{m}|=4\Rightarrow\sqrt{p^2+q^2+r^2}=4\Rightarrow p^2+q^2+r^2=16.$$

$$\cos 45^\circ=\frac{\begin{pmatrix}p\\q\\r\end{pmatrix}\cdot\begin{pmatrix}0\\1\\0\end{pmatrix}}{\sqrt{p^2+q^2+r^2}\times\sqrt{0^2+1^2+0^2}}.$$

$$\Rightarrow\frac{\sqrt{2}}{2}=\frac{p\times 0+q\times 1+r\times 0}{4\times 1}\Rightarrow\frac{\sqrt{2}}{2}=\frac{q}{4}\Rightarrow q=2\sqrt{2}.$$

$$\therefore p^2+(2\sqrt{2})^2+r^2=16\Rightarrow p^2+r^2=8.$$

Now solve the simultaneous equations $r=-3p$ and $p^2+r^2=8$:

$$p^2+(-3p)^2=8\Rightarrow p^2+9p^2=8\Rightarrow 10p^2=8\Rightarrow p^2=\frac{4}{5}\Rightarrow p=\pm\frac{2\sqrt{5}}{5}.$$

As p is a positive constant, $p=\frac{2\sqrt{5}}{5}$.

$$\therefore r=-3\left(\frac{2\sqrt{5}}{5}\right)=-\frac{6\sqrt{5}}{5}.$$

So $\mathbf{m}=\frac{2\sqrt{5}}{5}\mathbf{i}+2\sqrt{2}\mathbf{j}-\frac{6\sqrt{5}}{5}\mathbf{k}$.

25)

If $\mathbf{p}$ lies in the plane, then

$$\begin{pmatrix}12\\-4\\0\end{pmatrix}\cdot\begin{pmatrix}a\\b\\c\end{pmatrix}=0\Rightarrow 12a-4b=0\Rightarrow b=3a.$$

$$|\mathbf{p}|=6\sqrt{10}\Rightarrow\sqrt{a^2+b^2+c^2}=6\sqrt{10}\Rightarrow a^2+b^2+c^2=360.$$

$$\cos 30^\circ=\frac{\begin{pmatrix}a\\b\\c\end{pmatrix}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}}{\sqrt{a^2+b^2+c^2}\times\sqrt{0^2+0^2+1^2}}.$$

$$\Rightarrow\frac{\sqrt{3}}{2}=\frac{a\times 0+b\times 0+c\times 1}{6\sqrt{10}\times 1}\Rightarrow\frac{\sqrt{3}}{2}=\frac{c}{6\sqrt{10}}\Rightarrow c=3\sqrt{30}.$$

$$\therefore a^2+b^2+(3\sqrt{30})^2=360\Rightarrow a^2+b^2=90.$$

Now solve the simultaneous equations $b=3a$ and $a^2+b^2=90$:

$$a^2+(3a)^2=90\Rightarrow a^2+9a^2=90\Rightarrow 10a^2=90\Rightarrow a^2=9\Rightarrow a=\pm 3.$$

As a is a positive constant, $a=3$.

$$\therefore b=3(3)=9.$$

So $\mathbf{p}=3\mathbf{i}+9\mathbf{j}+3\sqrt{30}\mathbf{k}$.

26)

Let $\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$. If $\mathbf{v}$ is a unit vector then

$$\sqrt{a^2 + b^2 + c^2} = 1 \Rightarrow a^2 + b^2 + c^2 = 1.$$

If $\mathbf{v}$ is parallel to the plane $x - 2y = 5$ then:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} = 0 \Rightarrow a - 2b = 0 \Rightarrow b = \frac{1}{2}a.$$

$$\cos 45^\circ = \frac{\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 8 \\ -3 \\ 5 \end{pmatrix}}{\sqrt{a^2 + b^2 + c^2} \times \sqrt{8^2 + (-3)^2 + 5^2}}.$$

$$\Rightarrow \frac{\sqrt{2}}{2} = \frac{a \times 8 + b \times (-3) + c \times 5}{1 \times \sqrt{98}}.$$

$$\Rightarrow \frac{\sqrt{2}}{2} = \frac{8a - 3b + 5c}{7\sqrt{2}} \Rightarrow 8a - 3b + 5c = 7.$$

$$8a - 3\left(\frac{1}{2}a\right) + 5c = 7 \Rightarrow 5c = 7 - \frac{13}{2}a \Rightarrow c = \frac{7}{5} - \frac{13}{10}a.$$

$$a^2 + \left(\frac{1}{2}a\right)^2 + \left(\frac{7}{5} - \frac{13}{10}a\right)^2 = 1.$$

$$\Rightarrow a^2 + \frac{1}{4}a^2 + \frac{169}{100}a^2 - \frac{91}{25}a + \frac{49}{25} = 1 \Rightarrow \frac{147}{50}a^2 - \frac{91}{25}a + \frac{24}{25} = 0.$$

$$\Rightarrow 147a^2 - 182a + 48 = 0 \Rightarrow (21a - 8)(7a - 6) = 0.$$

$$\Rightarrow a = \frac{8}{21} \text{ or } a = \frac{6}{7}.$$

If $a = \frac{8}{21}$ then $b = \frac{4}{21}$ and $c = \frac{19}{21}$, so $\mathbf{v} = \frac{8}{21}\mathbf{i} + \frac{4}{21}\mathbf{j} + \frac{19}{21}\mathbf{k}$. If $a = \frac{6}{7}$ then $b = \frac{3}{7}$ and $c = \frac{2}{7}$, so $\mathbf{v} = \frac{6}{7}\mathbf{i} + \frac{3}{7}\mathbf{j} + \frac{2}{7}\mathbf{k}$.

27 a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 9 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ -5 \end{pmatrix} = \begin{pmatrix} 7 \\ -2 \\ 5 \end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} -7 \\ 3 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ -5 \end{pmatrix} = \begin{pmatrix} -9 \\ 0 \\ 7 \end{pmatrix}.$$

27 b)

$$\begin{pmatrix} 7 \\ -2 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ 47 \\ 9 \end{pmatrix} = 7 \times 7 + (-2) \times 47 + 5 \times 9 = 49 - 94 + 45 = 0.$$

$$\begin{pmatrix} -9 \\ 0 \\ 7 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ 47 \\ 9 \end{pmatrix} = (-9) \times 7 + 0 \times 47 + 7 \times 9 = -63 + 0 + 63 = 0.$$

$\therefore$ both $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are perpendicular to $7\mathbf{i} + 47\mathbf{j} + 9\mathbf{k}$.

27 c)

The plane has equation $7x + 47y + 9z = d$.

$$d = 7(2) + 47(3) + 9(-5) = 110.$$

$$\therefore 7x + 47y + 9z = 110.$$

28 a)

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \begin{pmatrix} 3 \\ 2 \\ 8 \end{pmatrix} - \begin{pmatrix} -1 \\ 2 \\ 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix}.$$

$$\overrightarrow{PR} = \overrightarrow{OR} - \overrightarrow{OP} = \begin{pmatrix} 4 \\ -3 \\ -5 \end{pmatrix} - \begin{pmatrix} -1 \\ 2 \\ 6 \end{pmatrix} = \begin{pmatrix} 5 \\ -5 \\ -11 \end{pmatrix}.$$

28 b)

$$\begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 27 \\ -10 \end{pmatrix} = 4 \times 5 + 0 \times 27 + 2 \times (-10) = 20 + 0 - 20 = 0.$$

$$\begin{pmatrix} 5 \\ -5 \\ -11 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 27 \\ -10 \end{pmatrix} = 5 \times 5 + (-5) \times 27 + (-11) \times (-10) = 25 - 135 + 110 = 0.$$

$\therefore$ both $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ are perpendicular to $5\mathbf{i} + 27\mathbf{j} - 10\mathbf{k}$.

28 c)

The plane has equation $5x + 27y - 10z = d$.

$$d = 5(-1) + 27(2) - 10(6) = -11.$$

$$\therefore 5x + 27y - 10z = -11.$$

29 a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 6 \\ -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix}.$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 2 \\ -2 & 1 & -1 \end{vmatrix}.$$

$$= ((-1) \times (-1) - 2 \times 1)\mathbf{i} - (2 \times (-1) - 2 \times (-2))\mathbf{j} + (2 \times 1 - (-1) \times (-2))\mathbf{k}.$$

$$= -\mathbf{i} - 2\mathbf{j}.$$

29 b)

The plane has equation $-x - 2y = d$.

$$d = -(4) - 2(0) + 0(-2) = -4.$$

$$\therefore -x - 2y = -4.$$

$$\Rightarrow x + 2y = 4.$$

30)

Let A, B and C be $(2, 1, 1)$, $(0, 1, 3)$ and $(1, 0, 4)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix}.$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix} \times \begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 0 & 2 \\ -1 & -1 & 3 \end{vmatrix}.$$

$$= (0 \times 3 - 2 \times (-1))\mathbf{i} - ((-2) \times 3 - 2 \times (-1))\mathbf{j} + ((-2) \times (-1) - 0 \times (-1))\mathbf{k}.$$

$$= 2\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}.$$

The plane has equation $2x + 4y + 2z = d$.

$$d = 2(2) + 4(1) + 2(1) = 10.$$

$$\therefore 2x + 4y + 2z = 10.$$

$$\Rightarrow x + 2y + z = 5.$$

31 a)

Let A, B and C be $(1, 0, 1)$, $(1, 1, 1)$ and $(0, 1, 0)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}.$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 0 \\ -1 & 1 & -1 \end{vmatrix}.$$

$$= (1 \times (-1) - 0 \times 1)\mathbf{i} - (0 \times (-1) - 0 \times (-1))\mathbf{j} + (0 \times 1 - 1 \times (-1))\mathbf{k}.$$

$$= -\mathbf{i} + \mathbf{k}.$$

The plane has equation $-x + z = d$.

$$d = -1(1) + 0(0) + 1(1) = 0.$$

$$\therefore -x + z = 0.$$

31 b)

Let A, B and C be $(2,1,1)$, $(-1,1,2)$ and $(3,-1,-1)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ -1 \\ -1 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix}.$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & 0 & 1 \\ 1 & -2 & -2 \end{vmatrix}.$$

$$= (0 \times (-2) - 1 \times (-2))\mathbf{i} - ((-3) \times (-2) - 1 \times 1)\mathbf{j} + ((-3) \times (-2) - 0 \times 1)\mathbf{k}.$$

$$= 2\mathbf{i} - 5\mathbf{j} + 6\mathbf{k}.$$

The plane has equation $2x - 5y + 6z = d$.

$$d = 2(2) - 5(1) + 6(1) = 5.$$

$$\therefore 2x - 5y + 6z = 5.$$

31 c)

Let A, B and C be $(-1,4,5)$, $(3,2,-5)$ and $(4,2,-3)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ -10 \end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \\ -8 \end{pmatrix}.$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 4 \\ -2 \\ -10 \end{pmatrix} \times \begin{pmatrix} 5 \\ -2 \\ -8 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -2 & -10 \\ 5 & -2 & -8 \end{vmatrix}.$$

$$= ((-2) \times (-8) - (-10) \times (-2))\mathbf{i} - (4 \times (-8) - (-10) \times 5)\mathbf{j} + (4 \times (-2) - (-2) \times 5)\mathbf{k}.$$

$$= -4\mathbf{i} - 18\mathbf{j} + 2\mathbf{k}.$$

The plane has equation $-4x - 18y + 2z = d$.

$$d = -4(-1) - 18(4) + 2(5) = -58.$$

$$\therefore -4x - 18y + 2z = -58.$$

$$\Rightarrow 2x + 9y - z = 29.$$

31 d)

Let A, B and C be $(10, -20, 30)$, $(-20, 40, 50)$ and $(0, 0, -10)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -20 \\ 40 \\ 50 \end{pmatrix} - \begin{pmatrix} 10 \\ -20 \\ 30 \end{pmatrix} = \begin{pmatrix} -30 \\ 60 \\ 20 \end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 0 \\ 0 \\ -10 \end{pmatrix} - \begin{pmatrix} 10 \\ -20 \\ 30 \end{pmatrix} = \begin{pmatrix} -10 \\ 20 \\ -40 \end{pmatrix}.$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} -30 \\ 60 \\ 20 \end{pmatrix} \times \begin{pmatrix} -10 \\ 20 \\ -40 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -30 & 60 & 20 \\ -10 & 20 & -40 \end{vmatrix}.$$

$$= (60 \times (-40) - 20 \times 20)\mathbf{i} - ((-30) \times (-40) - 20 \times (-10))\mathbf{j} + ((-30) \times 20 - 60 \times (-10))\mathbf{k}.$$

$$= -2800\mathbf{i} - 1400\mathbf{j}.$$

The plane has equation $-2800x - 1400y = d$.

$$d = -2800(10) - 1400(-20) = 0.$$

$$\therefore -2800x - 1400y = 0.$$

$$\Rightarrow 2x + y = 0.$$

32 a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 4 \\ 1 \\ -2 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 2 \\ 0 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 11 \end{pmatrix}.$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} \times \begin{pmatrix} -1 \\ -2 \\ 11 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 3 \\ -1 & -2 & 11 \end{vmatrix}.$$

$$= ((-1) \times 11 - 3 \times (-2))\mathbf{i} - (1 \times 11 - 3 \times (-1))\mathbf{j} + (1 \times (-2) - (-1) \times (-1))\mathbf{k}.$$

$$= -5\mathbf{i} - 14\mathbf{j} - 3\mathbf{k}.$$

The plane has equation $-5x - 14y - 3z = d$.

$$d = -5(3) - 14(2) - 3(-5) = -28.$$

$$\therefore -5x - 14y - 3z = -28.$$

$$\Rightarrow 5x + 14y + 3z = 28.$$

$$5(k) + 14(1) + 3(2) = 28 \Rightarrow 5k + 20 = 28 \Rightarrow k = \frac{8}{5}.$$

32 b)

$$5(3) + 14(k) + 3(2) = 28 \Rightarrow 21 + 14k = 28 \Rightarrow k = \frac{1}{2}.$$

32 c)

$$5(0) + 14(5) + 3(k) = 28 \Rightarrow 70 + 3k = 28 \Rightarrow k = -14.$$

33)

$$\begin{pmatrix} 2 & -3 & k \\ 4 & 1 & -10 \\ -2 & -11 & 5 \end{pmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix} = \begin{pmatrix} 2p - 3q + kr \\ 4p + q - 10r \\ -2p - 11q + 5r \end{pmatrix}.$$

Substitute these into the equation of the plane:

$$3(2p - 3q + kr) - 2(4p + q - 10r) - (-2p - 11q + 5r) = 0.$$

$$\Rightarrow 6p - 9q + 3kr - 8p - 2q + 20r + 2p + 11q - 5r = 0.$$

$$\Rightarrow 3kr + 15r = 0.$$

$$\Rightarrow k = -5.$$

34 a)

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}.$$

$$\overrightarrow{AD} \times \overrightarrow{AB} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 1 \\ 2 & -1 & -2 \end{vmatrix}.$$

$$= (2 \times (-2) - 1 \times (-1))\mathbf{i} - (1 \times (-2) - 1 \times 2)\mathbf{j} + (1 \times (-1) - 2 \times 2)\mathbf{k}.$$

$$= -3\mathbf{i} + 4\mathbf{j} - 5\mathbf{k}.$$

The plane has equation $-3x + 4y - 5z = d$.

$$d = -3(0) + 4(1) - 5(3) = -11.$$

$$\therefore -3x + 4y - 5z = -11.$$

$$\Rightarrow 3x - 4y + 5z = 11.$$

34 b)

$$3(4) - 4(1) + 5(k) = 11 \Rightarrow 8 + 5k = 11 \Rightarrow k = \frac{3}{5}.$$

34 c)

$$\cos\theta = \frac{\begin{pmatrix}3\\-4\\5\end{pmatrix}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}}{\sqrt{3^2+(-4)^2+5^2}\times\sqrt{0^2+0^2+1^2}}.$$

$$\Rightarrow \cos\theta = \frac{3\times 0+(-4)\times 0+5\times 1}{\sqrt{50}\times\sqrt{1}}.$$

$$\Rightarrow \cos\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = 45^\circ.$$

Acute Angle $= 45^\circ$.

35)

First find Π_1:

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}2\\0\\-2\end{pmatrix} - \begin{pmatrix}1\\2\\-1\end{pmatrix} = \begin{pmatrix}1\\-2\\-1\end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix}0\\1\\1\end{pmatrix} - \begin{pmatrix}1\\2\\-1\end{pmatrix} = \begin{pmatrix}-1\\-1\\2\end{pmatrix}.$$

$$\overrightarrow{AB}\times\overrightarrow{AC} = \begin{pmatrix}1\\-2\\-1\end{pmatrix}\times\begin{pmatrix}-1\\-1\\2\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\1 & -2 & -1\\-1 & -1 & 2\end{vmatrix}.$$

$$= ((-2)\times 2-(-1)\times(-1))\mathbf{i} - (1\times 2-(-1)\times(-1))\mathbf{j} + (1\times(-1)-(-2)\times(-1))\mathbf{k}.$$

$$= -5\mathbf{i} - \mathbf{j} - 3\mathbf{k}.$$

The plane has equation $-5x - y - 3z = d$.

$$d = -5(1) - (2) - 3(-1) = -4.$$

$$\therefore -5x - y - 3z = -4.$$

$$\Rightarrow \Pi_1 : 5x + y + 3z = 4.$$

Now find Π_2:

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}2\\0\\-2\end{pmatrix} - \begin{pmatrix}1\\2\\-1\end{pmatrix} = \begin{pmatrix}1\\-2\\-1\end{pmatrix}.$$

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \begin{pmatrix}3\\-2\\1\end{pmatrix} - \begin{pmatrix}1\\2\\-1\end{pmatrix} = \begin{pmatrix}2\\-4\\2\end{pmatrix}.$$

$$\overrightarrow{AB}\times\overrightarrow{AC} = \begin{pmatrix}1\\-2\\-1\end{pmatrix}\times\begin{pmatrix}2\\-4\\2\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\1 & -2 & -1\\2 & -4 & 2\end{vmatrix}.$$

$$= ((-2)\times 2-(-1)\times(-4))\mathbf{i} - (1\times 2-(-1)\times 2)\mathbf{j} + (1\times(-4)-(-2)\times 2)\mathbf{k}.$$

$$= -8\mathbf{i} - 4\mathbf{j}.$$

The plane has equation $-8x - 4y = d$.

$$d = -8(1) - 4(2) = -16.$$

$$\therefore -8x - 4y = -16.$$
$$\Rightarrow \Pi_2 : 2x + y = 4.$$

Now find the acute angle between Π_1 and Π_2:

$$\cos\theta = \frac{\begin{pmatrix}5\\1\\3\end{pmatrix} \cdot \begin{pmatrix}2\\1\\0\end{pmatrix}}{\sqrt{5^2+1^2+3^2} \times \sqrt{2^2+1^2+0^2}}.$$
$$\Rightarrow \cos\theta = \frac{5\times 2 + 1\times 1 + 3\times 0}{\sqrt{35}\sqrt{5}}.$$
$$\Rightarrow \cos\theta = \frac{11}{5\sqrt{7}} = \frac{11\sqrt{7}}{35} \Rightarrow \theta = 33.744\ldots^\circ.$$

Acute Angle $= 33.7^\circ$ (1 DP).

36 a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}\frac{\sqrt{15}}{2}\\0\\-\frac{\sqrt{5}}{2}\end{pmatrix} - \begin{pmatrix}0\\0\\\sqrt{5}\end{pmatrix} = \begin{pmatrix}\frac{\sqrt{15}}{2}\\0\\-\frac{3\sqrt{5}}{2}\end{pmatrix}.$$
$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \begin{pmatrix}0\\\sqrt{10}\\0\end{pmatrix} - \begin{pmatrix}0\\0\\\sqrt{5}\end{pmatrix} = \begin{pmatrix}0\\\sqrt{10}\\-\sqrt{5}\end{pmatrix}.$$
$$\overrightarrow{AB} \times \overrightarrow{AD} = \begin{pmatrix}\frac{\sqrt{15}}{2}\\0\\-\frac{3\sqrt{5}}{2}\end{pmatrix} \times \begin{pmatrix}0\\\sqrt{10}\\-\sqrt{5}\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\\frac{\sqrt{15}}{2} & 0 & -\frac{3\sqrt{5}}{2}\\0 & \sqrt{10} & -\sqrt{5}\end{vmatrix}.$$
$$= \left(0\times(-\sqrt{5}) - \left(-\frac{3\sqrt{5}}{2}\right)\times\sqrt{10}\right)\mathbf{i}$$
$$- \left(\frac{\sqrt{15}}{2}\times(-\sqrt{5}) - \left(-\frac{3\sqrt{5}}{2}\right)\times 0\right)\mathbf{j}$$
$$+ \left(\frac{\sqrt{15}}{2}\times\sqrt{10} - 0\times 0\right)\mathbf{k}.$$
$$= \frac{15\sqrt{2}}{2}\mathbf{i} + \frac{5\sqrt{3}}{2}\mathbf{j} + \frac{5\sqrt{6}}{2}\mathbf{k}.$$

The plane has equation

$$\frac{15\sqrt{2}}{2}x + \frac{5\sqrt{3}}{2}y + \frac{5\sqrt{6}}{2}z = d.$$
$$d = \frac{15\sqrt{2}}{2}(0) + \frac{5\sqrt{3}}{2}(0) + \frac{5\sqrt{6}}{2}(\sqrt{5}) = \frac{5\sqrt{30}}{2}.$$
$$\therefore \frac{15\sqrt{2}}{2}x + \frac{5\sqrt{3}}{2}y + \frac{5\sqrt{6}}{2}z = \frac{5\sqrt{30}}{2}.$$
$$\Rightarrow 3\sqrt{2}x + \sqrt{3}y + \sqrt{6}z = \sqrt{30}.$$

36 b)

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} -\frac{\sqrt{15}}{2} \\ 0 \\ -\frac{\sqrt{5}}{2} \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ \sqrt{5} \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{15}}{2} \\ 0 \\ -\frac{3\sqrt{5}}{2} \end{pmatrix}.$$

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = \begin{pmatrix} 0 \\ \sqrt{10} \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ \sqrt{5} \end{pmatrix} = \begin{pmatrix} 0 \\ \sqrt{10} \\ -\sqrt{5} \end{pmatrix}.$$

$$\overrightarrow{AC} \times \overrightarrow{AD} = \begin{pmatrix} -\frac{\sqrt{15}}{2} \\ 0 \\ -\frac{3\sqrt{5}}{2} \end{pmatrix} \times \begin{pmatrix} 0 \\ \sqrt{10} \\ -\sqrt{5} \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\frac{\sqrt{15}}{2} & 0 & -\frac{3\sqrt{5}}{2} \\ 0 & \sqrt{10} & -\sqrt{5} \end{vmatrix}.$$

$$= \left(0 \times (-\sqrt{5}) - \left(-\frac{3\sqrt{5}}{2}\right) \times \sqrt{10}\right)\mathbf{i}$$
$$- \left(\left(-\frac{\sqrt{15}}{2}\right) \times (-\sqrt{5}) - \left(-\frac{3\sqrt{5}}{2}\right) \times 0\right)\mathbf{j}$$
$$+ \left(\left(-\frac{\sqrt{15}}{2}\right) \times \sqrt{10} - 0 \times 0\right)\mathbf{k}.$$

$$= \frac{15\sqrt{2}}{2}\mathbf{i} - \frac{5\sqrt{3}}{2}\mathbf{j} - \frac{5\sqrt{6}}{2}\mathbf{k}.$$

The plane has equation

$$\frac{15\sqrt{2}}{2}x - \frac{5\sqrt{3}}{2}y - \frac{5\sqrt{6}}{2}z = d.$$

$$d = \frac{15\sqrt{2}}{2}(0) - \frac{5\sqrt{3}}{2}(0) - \frac{5\sqrt{6}}{2}(\sqrt{5}) = -\frac{5\sqrt{30}}{2}.$$

$$\therefore \frac{15\sqrt{2}}{2}x - \frac{5\sqrt{3}}{2}y - \frac{5\sqrt{6}}{2}z = -\frac{5\sqrt{30}}{2}.$$

$$\Rightarrow 3\sqrt{2}x - \sqrt{3}y - \sqrt{6}z = -\sqrt{30}.$$

36 c)

$$\cos\theta = \frac{\begin{pmatrix} 3\sqrt{2} \\ \sqrt{3} \\ \sqrt{6} \end{pmatrix} \cdot \begin{pmatrix} 3\sqrt{2} \\ -\sqrt{3} \\ -\sqrt{6} \end{pmatrix}}{\sqrt{(3\sqrt{2})^2 + (\sqrt{3})^2 + (\sqrt{6})^2} \times \sqrt{(3\sqrt{2})^2 + (-\sqrt{3})^2 + (-\sqrt{6})^2}}.$$

$$\Rightarrow \cos\theta = \frac{3\sqrt{2} \times 3\sqrt{2} + \sqrt{3} \times (-\sqrt{3}) + \sqrt{6} \times (-\sqrt{6})}{\sqrt{27} \times \sqrt{27}}.$$

$$\Rightarrow \cos\theta = \frac{9}{27} = \frac{1}{3} \Rightarrow \theta = 70.528\ldots^\circ.$$

Acute angle is 70.5° (1 DP).